Talank Baral

Ph.D. Student in Computer Science

Department of Computer Science, George Mason University • Falls Church, VA 22046, USA

tbaral@gmu.edu talank.com.np github.com/talank in/tlnkb 0009-0000-9705-0693

Research Interests

My research improves the speed and reliability of software testing and continuous integration (CI), combining empirical studies with program analysis and large language models (LLMs). I work on reproducing and repairing flaky tests, optimizing test execution and CI pipelines, and understanding how AI coding agents test the code they write. My research is motivated by two years of industry experience maintaining slow, brittle test pipelines.

Software Testing · Continuous Integration · Flaky Tests · Regression Test Optimization · LLMs for Software Engineering · Agentic Software Engineering

Education

2022–Present **Ph.D. in Computer Science** — Fairfax, VA, USA
George Mason University
GPA: 3.98/4.00 • Advisor: Wing Lam; expected graduation May 2027

2015–2019 **B.E. in Software Engineering** — Pokhara, Nepal
Gandaki College of Engineering and Science, Pokhara University
GPA: 3.67/4.00

Publications

Conference Papers

[C3] ASE '26 Shanto Rahman, **Talank Baral**, August Shi, and Wing Lam. **TDRepro: A Neurosymbolic Approach to Reproducing Timing-Dependent Flaky Test Failures**. *41st IEEE/ACM International Conference on Automated Software Engineering (ASE 2026)*, pp. 675–686, Munich, Germany, 2026.
ACM SIGSOFT Distinguished Paper Award
Acceptance rate: 8.82% direct acceptance [DOI] [PDF]

[C2] ICSME '26 Atish Kumar Dipongkor, **Talank Baral**, Wing Lam, and Kevin Moran. **Test Coverage Analysis of Agentic Pull Requests**. *42nd International Conference on Software Maintenance and Evolution (ICSME 2026)*, Vision and Emerging Results Track, Benevento, Italy, 2026.
Acceptance rate: 39% [arXiv] [Code] [Artifact]

[C1] ASE '23 **Talank Baral**, Shanto Rahman, Bala Naren Chanumolu, Basak Balci, Tuna Tuncer, August Shi, and Wing Lam. **Optimizing Continuous Development By Detecting and Preventing Unnecessary Content Generation**. *38th IEEE/ACM International Conference on Automated Software Engineering (ASE 2023)*, pp. 901–913, Kirchberg, Luxembourg, 2023.
Acceptance rate: 21% (134/630) [PDF] [Slides]

Tool Demonstrations

[D1] ICSE '25 Demo — **Talank Baral**, Emirhan Oğul, Shanto Rahman, August Shi, and Wing Lam. **OptCD: Optimizing Continuous Development**. *47th IEEE/ACM International Conference on Software Engineering (ICSE 2025)*, Demonstrations Track, pp. 45–48, Ottawa, ON, Canada, 2025. [PDF] [Slides]

Under Review

[U1] **Talank Baral**, Abdelrahman Baz, Atish Kumar Dipongkor, August Shi, Kevin Moran, and Wing Lam. **Using Efficient Test Orders for Regression Testing**. *Under review*.

Honors & Awards

Oct 2026 — **ACM SIGSOFT Distinguished Paper Award**, ASE '26
TDRepro: A Neurosymbolic Approach to Reproducing Timing-Dependent Flaky Test Failures

Aug 2023 — **SIGSOFT CAPS Travel Grant**, ACM SIGSOFT
Awarded $1,200 to attend ASE 2023 (declined due to visa constraints)

May 2023 — **Summer Research Award**, Department of Computer Science, George Mason University

2015–2019 — **Undergraduate Academic Excellence Award**, Gandaki College of Engineering and Science, Pokhara University
Merit-based award

Feb 2018 — **First Place, Project Expo**, 7th GCES IT Expo, Pokhara University

Jun 2012 — **National Talent Identification Award (First Place Nationwide)**, Government of Nepal
For designing a seed plantation machine

Experience

Aug 2022–Present — **Graduate Research/Teaching Assistant** — Fairfax, VA, USA
Department of Computer Science, George Mason University

- Reproduced timing-dependent flaky test failures with a neurosymbolic LLM pipeline (TDRepro; ASE '26 Distinguished Paper)
- Detected and prevented unnecessary content generation in CI builds (OptCD; ASE '23 and ICSE '25 Demo)
- Studied the test coverage of pull requests authored by AI coding agents (ICSME '26)
- Developed techniques for efficient test orders in regression testing
- Teaching assistant for graduate and undergraduate computer science courses

Jan 2020–Aug 2022 — **Software Developer** — Pokhara, Nepal
JankariTech

- Set up test automation, CI pipelines, and acceptance tests for client projects
- Wrote and maintained end-to-end and API test suites for ownCloud (Germany), an open-source content collaboration platform

Teaching

Spring 2026 — **CS 691**: Synergies of LLMs and Software Engineering — *Graduate Teaching Assistant*

Fall 2022 — **CS 262**: Introduction to Low-Level Programming — *Graduate Teaching Assistant*

Invited & Conference Talks

May 2025	*OptCD: Optimizing Continuous Development* ICSE 2025 (Demonstrations Track) • Conference Talk • Ottawa, ON, Canada
Jan 2024	*Optimizing Continuous Development By Detecting and Preventing Unnecessary Content Generation* ESSEC 2024 • Colloquium Talk • William & Mary, Williamsburg, VA, USA
Jul 2023	*Optimizing Continuous Development* Microsoft, Redmond Campus • Industry Talk • Redmond, WA, USA
Jul 2023	*Optimizing Continuous Development and Predicting Test Information in Software Development* ISSTA 2023 • Doctoral Symposium • Seattle, WA, USA

Open-Source Contributions

- **International Dataset of Flaky Tests (IDoFT)**: Detected and fixed flaky tests in open-source Java projects
- **CI build optimization**: Submitted 25 pull requests to open-source projects that disable generation of unnecessary content in CI builds
- **ownCloud**: Automated tests for the ownCloud web and desktop clients
- **pact-cplusplus**: Fixed compilation issues in version 0.1.0

Technical Skills

Languages	Python, Java, Bash, C, C++, JavaScript, SQL
Testing	JUnit, Maven, Selenium, Cypress, Nightwatch.js, Jest, Puppeteer, Behat
CI/CD	GitHub Actions, GitLab CI, Travis CI, Drone, Docker